

On the Formal Foundation of Boundary 1

A Lawvere-Yanofsky Proof That Verification Architectures Cannot Attest Their Own Consistency

Thorben Liebig
Cyber Resilience Architect
CISM, CRISC (ISACA Member 2226146)
ORCID: 0009-0006-2785-7911

May 16, 2026

Abstract

A verification architecture that evaluates assertions entering binding decision chains and that can express statements about its own verification procedures exhibits the structural self-reference condition identified by the diagonal arguments of Goedel, Tarski, Cantor, and Turing. This note grounds the architectural conclusion (that the record of inscription, verification, and attribution must stand outside the verification architecture) in the Lawvere-Yanofsky generalisation of diagonal arguments. Lawvere (1969) showed that the central diagonal results of twentieth-century logic are instances of a single fixed-point theorem in Cartesian closed categories. Yanofsky (2003) gave an accessible set-theoretic exposition of the same universal scheme. The scheme requires no assumption that the target system is a recursively enumerable formal system with mechanical inference rules. It requires three structural conditions: a domain with products, a self-referential encoding capacity, and a negation-like function without fixed points.

This note proceeds in two steps. First, it states and proves the self-attestation bound for any verification architecture satisfying the three conditions, at the level of generality the proof supports (Sections 3–5). Second, it shows that institutional verification architectures satisfy the conditions by constitutive necessity, and are therefore an instance of the general result (Section 6).

The Problem

A companion paper derives Boundary 1 from Goedel’s second incompleteness theorem¹ applied to the verification architecture as its proper target. Boundary 1 is the requirement that the record must stand outside the verification architecture. The derivation proceeds through five steps: verification architectures that govern assertions entering binding decision chains must protect their epistemic integrity; the structure that

¹ Kurt Goedel, “Ueber formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I,” *Monatshefte fuer Mathematik und Physik* 38 (1931): 173–198.

makes this protection operative is an axiom-based knowledge space; that knowledge space is sufficiently expressive to fall under the second incompleteness theorem; therefore the architecture cannot attest its own consistency from within; therefore the record must stand outside.

A philosophical reviewer will raise the objection that verification architectures, particularly institutional ones, are not formal axiomatic systems in the strict Hilbert-Goedel sense. They operate on mixed materials, their inference rules are interpretive, and their corpora of authority are open-ended.

This note makes the response precise. It shows that the architectural conclusion follows from a generalisation of Goedel's theorem that operates at a level of abstraction above the distinction between strict-formal and institutional-formal systems. The generalisation is Lawvere's fixed-point theorem (1969), as explicated by Yanofsky (2003). The proof is stated for verification architectures in general; the application to institutional verification architectures follows as an instantiation.

The Lawvere-Yanofsky Generalisation

In 1969, F. William Lawvere proved a fixed-point theorem in Cartesian closed categories that unified the diagonal arguments underlying many of the central results of twentieth-century logic.² Yanofsky, in his accessible exposition, characterised the scope of the result: *many self-referential paradoxes, incompleteness theorems and fixed point theorems fall out of the same simple scheme.*³ In a 2007 interview, Lawvere himself described his result as having *demystified the incompleteness theorem of Goedel and the truth-definition theory of Tarski by showing that both are consequences of some very simple algebra in the Cartesian-closed setting.*⁴

Yanofsky (2003) gave an accessible set-theoretic exposition of the same scheme, demonstrating that Cantor's theorem, Russell's paradox, Tarski's undefinability, Goedel's first incompleteness theorem, the Goedel-Rosser theorem, the halting problem, and Rice's theorem are all instances of a single diagonal argument.⁵

The abstract scheme, stated in set-theoretic terms following Yanofsky, is:

Theorem 1 (Lawvere-Yanofsky Diagonal Lemma). *Let A be a set, let B be a set with at least two elements, and let $\varphi: A \rightarrow B^A$ be a surjection (i.e., every function from A to*

² F. William Lawvere, "Diagonal Arguments and Cartesian Closed Categories," in *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92 (Berlin: Springer, 1969), 134–145; reprinted in *Reprints in Theory and Applications of Categories* 15 (2006), 1–13.

³ Noson S. Yanofsky, "A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points," *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386, at 362 (abstract).

⁴ Quoted in the author commentary to the 2006 reprint of Lawvere (1969), p. 2.

⁵ Noson S. Yanofsky, "A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points," *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386.

B is in the range of φ). Then every function $g: B \rightarrow B$ has a fixed point: there exists $b \in B$ such that $g(b) = b$.

Proof. Define $f: A \rightarrow B$ by $f(a) = g(\varphi(a)(a))$ for all $a \in A$. Since φ is surjective, there exists $a_0 \in A$ such that $\varphi(a_0) = f$. Then:

$$f(a_0) = g(\varphi(a_0)(a_0)) = g(f(a_0))$$

Setting $b = f(a_0)$, we have $g(b) = b$. $\square$

The contrapositive is the diagonal argument in its universal form:

Corollary 2 (Universal Diagonal Argument). *Let A be a set, let B be a set, and let $g: B \rightarrow B$ be a function with no fixed point (i.e., $g(b) \neq b$ for all $b \in B$). Then no function $\varphi: A \rightarrow B^A$ is surjective.*

Remark 3. *Goedel's incompleteness theorem is the instance where A is the set of Goedel numbers, $B = \{\text{provable}, \text{unprovable}\}$, φ is the encoding that maps Goedel numbers to the provability status of propositions about Goedel numbers, and g is negation (swapping provable and unprovable). Since g has no fixed point in a consistent system (no proposition is both provable and unprovable), φ cannot be surjective: the system cannot fully encode its own provability relation.*

Remark 4. *Tarski's undefinability theorem is the instance where $B = \{\text{true}, \text{false}\}$, φ is the encoding that maps Goedel numbers to the truth-value assignments of propositions about Goedel numbers, and g is negation. Since g has no fixed point, the system cannot define its own truth predicate from within.*

Remark 5. *The scheme does not require that A be recursively enumerable, that φ be computable, or that the system be a formal system in any specific technical sense. It requires only that A , B , φ , and g exist as set-theoretic objects with the stated properties. This is the level of abstraction at which the general proof operates.*

Verification Architecture as a System

The proof of the self-attestation bound requires a definition of the target system at the level of generality the diagonal scheme supports. The definition identifies the minimal primitives; all other components are derived.

Definition 6 (Verification Architecture). *A verification architecture is a pair $\mathcal{V} = (A, C)$ where:*

- *A is a set of assertions that enter decision chains, closed under conjunction and disjunction (an algebra over its own verification statements);*

- C is the certification function with codomain $\{+, -\}$, where $+$ denotes admitted and $-$ denotes rejected. C is well-defined: for any input pair it returns exactly one value. The domain of C is specified below.

The following components are derived from (A, C) :

- The record R is the set of evaluations C has produced: $R = \{(a, v, C(a, v)) \mid C(a, v) \text{ has been evaluated}\}$. The record documents which assertions were examined, by which procedure, and with what result. Rejected assertions are part of the record.
- The verification procedures V are the image of C over the record: $V = C^R = \pi_2(R)$. V is not a free-standing primitive. The procedures that exist are the procedures that have been exercised and recorded. By the expressiveness condition below, $V \subseteq A$.

The domain of C is $A \times V$. Since $V \subseteq A$, the domain is a subset of $A \times A$. C evaluates assertions against verification procedures; it is not defined on arbitrary assertion-assertion pairs.

Condition 7 (Expressiveness). $\mathcal{V}$ is sufficiently expressive if A contains assertions about its own verification procedures. That is, for any $a \in A$ and any $v \in V$, the statement “ a was verified by v and admitted” is itself in A .

The Self-Attestation Bound

The bound applies to architectures claiming complete internal self-attestation. Architectures that make no such claim are not the target; they already operate with an external boundary.

Theorem 8 (Self-Attestation Bound). Let $\mathcal{V} = (A, C)$ be a verification architecture satisfying Condition 7. Let $B = \{+, -\}$. Then there is no surjective $\varphi: A \rightarrow B^A$.

Proof. The proof is a closed chain of four steps.

Step 1 (Structured domain). A is closed under conjunction and disjunction (by definition). The architecture can represent paired or jointly evaluated assertions, either by ordered-pair coding or by conjunction. This supplies the product-like structure required for the diagonal construction.

Step 2 (Self-referential encoding). By Condition 7, for any $a \in A$ and $v \in V = C^R$, the verification-consistency status of a under v is itself expressible in A . Therefore a function $\varphi: A \rightarrow B^A$ exists that encodes, for each assertion, the verification-consistency status the record assigns to every other assertion.

Step 3 (No fixed point). Let $g: B \rightarrow B$ be negation: $g(+) = -, g(-) = +$. Since C is well-defined (returns exactly one value), g has no fixed point on the two-element set B .

Step 4 (Diagonal argument). A has product-like pairing capacity (Step 1), $\varphi: A \rightarrow B^A$ exists (Step 2), g has no fixed point (Step 3). By Corollary 2, φ is not surjective. There exist verification-consistency assignments over the architecture's own assertions that φ does not encode. The architecture cannot fully resolve its own verification-consistency from within. $\square$

Corollary 9 (The Record Must Stand Outside). *For any verification architecture $\mathcal{V}$ satisfying Condition 7, the record R cannot be maintained by the verification architecture from within its own loop. The record must stand at a position external to the verification architecture.*

Proof. By Theorem 8, the architecture cannot fully resolve its own consistency. A record maintained from within the verification loop would itself be an assertion whose consistency status the architecture cannot internally attest. Placing the record outside the loop, in a form the architecture cannot modify from within, removes the self-referential encoding that generates the constraint. $\square$

Institutional Architectures as an Instance

The result of the preceding sections applies to any verification architecture satisfying Condition 7. This section shows that institutional verification architectures satisfy the condition by constitutive necessity.

Definition 10 (Institutional Verification Architecture). *An institutional verification architecture is a verification architecture $\mathcal{V} = (A, C)$ (Definition 6) where the assertions in A enter institutional decision chains (regulatory findings, audit opinions, compliance certifications, judicial determinations, and any other propositional content that may be relied upon in producing a binding decision).*

Proposition 11. *Any institutional verification architecture satisfies Condition 7.*

Proof. An institution that cannot state, as a matter of record, what was verified, by what procedure, and when, cannot be audited, cannot be held to account in adjudication, and cannot defend its decisions under regulatory inspection. Expressiveness is constitutive of the institution's capacity to bind. The condition is not optional for institutions that produce binding decisions; it is a consequence of their institutional character. $\square$

Corollary 12. *Institutional verification architectures fall within the scope of Theorem 8 and Corollary 9. No internal procedure of an institutional verification architecture can fully resolve the consistency of the institution's own verification operations. The record must stand outside.*

Proof. By Proposition 11, institutional verification architectures satisfy Condition 7. The result follows directly from Theorem 8 and Corollary 9. $\square$

Remark 13. *The general result (Theorem [8](#)) does not require that the system be:*

- *recursively enumerable (the axioms of authority may be open-ended; statutes change, regulations are amended, case law develops);*
- *mechanically decidable (the inference rules may be interpretive; legal reasoning involves judgment, not computation);*
- *finitely axiomatisable (the corpus of accepted assertions may grow without bound; institutional records accumulate over time);*
- *syntactically closed (the language in which assertions are formulated may evolve; regulatory terminology changes, professional standards are updated).*

Each of these properties distinguishes institutional verification architectures from strict Hilbert-Goedel systems. None of them affects the three conditions of the diagonal scheme. The product structure holds because A is closed under conjunction and disjunction, regardless of how assertions were generated. The self-referential encoding holds because expressiveness is a property of A , not of its formal closure. The absence of fixed points is a purely combinatorial fact about negation on a two-element set.

The Lawvere Bound

Definition 14 (The Lawvere Bound). *A system S falls within the Lawvere bound if it satisfies the three conditions of the diagonal scheme: structured domain with products, self-referential encoding capacity, and a negation-like function without fixed points. Any system within the Lawvere bound is subject to the self-attestation constraint of Theorem [8](#), regardless of whether the system is a strict Hilbert-Goedel formal system or not.*

Proposition 15. *Strict Hilbert-Goedel formal systems fall within the Lawvere bound (this is Goedel's original result, recognised as a special case by Lawvere 1969). Verification architectures satisfying Condition [7](#) fall within the Lawvere bound (this is Theorem [8](#)). Institutional verification architectures fall within the Lawvere bound (this is Corollary [12](#)). The architectural conclusion, that the record must stand outside the verification loop, holds for any system within the Lawvere bound, and is therefore independent of the differences between strict-formal, institutional-formal, and any other system class satisfying the three conditions.*

Proof. All three classes of systems satisfy the three conditions. The architectural conclusion is derived from the three conditions alone. The differences between the classes (recursive enumerability, mechanical decidability, finite axiomatisability, syntactic closure) do not appear in any premise of the derivation. The conclusion is therefore invariant under the differences.

The Volume Condition

The companion paper argues that LLMs are the volume condition under which the latent constraint identified by Theorem 8 became operationally binding. Before LLMs, the volume of assertions entering verification architectures was bounded by human production capacity. The self-attestation constraint applied throughout, but architectures could proceed informally, as if internal review constituted consistency attestation, because the volume did not force confrontation with the constraint.

LLMs collapsed the marginal cost of producing assertion-candidates. The volume at which plausible-looking assertions enter decision chains now exceeds the capacity of informal verification to absorb the latent constraint. The constraint did not change. The condition under which the constraint became non-ignorable did.

Theorem 8 and Corollary 9 together state the structural result. The volume condition states the empirical fact that has made the structural result operationally binding. Neither depends on the other: the structural result holds regardless of volume; the volume condition would create operational difficulty regardless of the structural result. Together, they establish that the architectural response (an external record, a metalanguage verification operation, a graduated verification regime, and a human gate) is both structurally required and operationally urgent.

Conclusion

The gap between strict Hilbert-Goedel systems and institutional verification architectures is real. The gap does not affect the architectural conclusion. The Lawvere-Yanofsky generalisation operates at a level of abstraction that encompasses both strict-formal and institutional-formal systems, provided both exhibit self-referential encoding capacity and consistency presumption. The general result (Theorem 8) is stated for any verification architecture satisfying the two conditions; the institutional application (Corollary 12) follows as an instance. The architectural conclusion, that the record must stand outside, follows from the generalised diagonal argument applied to its proper target, with no residual dependence on the strict-formal properties the institution does not possess.

References

- Kurt Goedel, “Ueber formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I,” *Monatshefte fuer Mathematik und Physik* 38 (1931): 173–198.
- F. William Lawvere, “Diagonal Arguments and Cartesian Closed Categories,” in *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92 (Berlin: Springer, 1969), 134–145; reprinted in *Reprints in Theory and Applications of Categories* 15 (2006), 1–13.

Noson S. Yanofsky, “A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points,” *Bulletin of Symbolic Logic* 9, no. 3 (2003): 362–386.

Alan M. Turing, “On Computable Numbers, with an Application to the Entscheidungsproblem,” *Proceedings of the London Mathematical Society* s2-42 (1936): 230–265.

Alfred Tarski, *Pojecie prawdy w jezykach nauk dedukcyjnych* (Warsaw: Nakladem Towarzystwa Naukowego Warszawskiego, 1933); German translation: “Der Wahrheitsbegriff in den formalisierten Sprachen,” *Studia Philosophica* 1 (1935): 261–405.

Declaration on the use of AI tools. *This paper was developed with the assistance of Claude (Anthropic, Claude Opus 4.6). The instrument was used for structural drafting, LaTeX formatting, editorial iteration, and bibliographic cross-referencing. All substantive claims, mathematical proofs, legal analysis, doctrinal positions, and architectural decisions are the author's. The instrument produced no assertion that entered the final text without human verification at the gate. The verification architecture described in this paper was applied to its own production.*